

Credit Card Fraud

How it impacts consumers trends in the current landscape and what everyone can do to prevent it

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What is Credit Card Fraud?

Credit card fraud is when an unauthorized person uses a card to make unauthorized purchases or withdrawals by using physical card theft card skimming devices or online information stealing to receive the card information (OCC 2025).

Two sides to the problem:

1. The impact of credit card fraud on consumers.
2. Trends in the credit card fraud landscape and what businesses can do to mitigate it as much as possible.

Card Fraud History



There were 503,450 reported cases in the first three quarters of 2025 which was a 54% growth over prior year.

Two types of credit card fraud:

1. New account [an identity thief uses a person's information to open a new card in that person's name]
2. Existing account [a thief uses your existing account information to make fraudulent purchases]

New account fraud accounts for 90% of all credit card fraud.

Card Fraud Repercussions



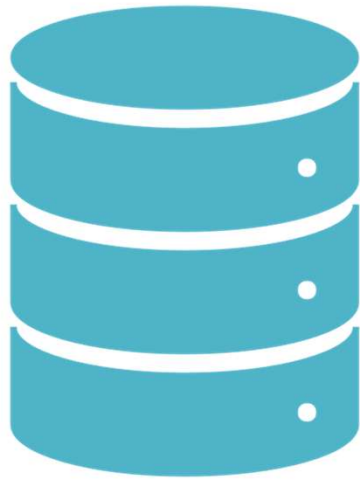
For those people that were found and sentenced for credit card fraud:

- 74.7% were men
- 55.5% had little to no criminal history
- 82.8% were United States citizens with an average age of 38 years old
- \$154,919 was the median loss
- 26 months was the average time in prison



Data Set Selection & Exploratory Data Analysis

Potential Data Sets



Two potential data sets were found both through Kaggle.

- The first is a synthetic data set with 1,000,000 records with 26 columns of data related to the transaction as well as the customer and account (Islam 2026).
- The second was a data set generated by Vesta's real world transaction data and created as a competition data set within Kaggle.
- *There were for the over 1 million rows 871 columns with column names as simple letter and number combinations. There was no data dictionary provided. While the data was already provided in test and training data sets the ambiguity of the column names made interpretation of the potential results more difficult.*

Having worked in the financial industry the synthetic data seemed to have the most relevant data elements that a bank would use in their fraud mitigation strategies.

Selected Data Set

Bank Transaction Fraud Detection Dataset (BTFDD)

Created with synthetic data and 26 columns

Transaction

- time
- date
- hour of day
- weekend indicator
- night indicator
- country
- city
- merchant category
- payment method
- device type
- distance from home (km)
- time since last transaction (hours)

- international indicator
- failed attempts count
- transaction amount
- number of previous transactions
- monthly transaction frequency

Customer

- age
- credit score
- age of account
- account balance
- pin changed recently

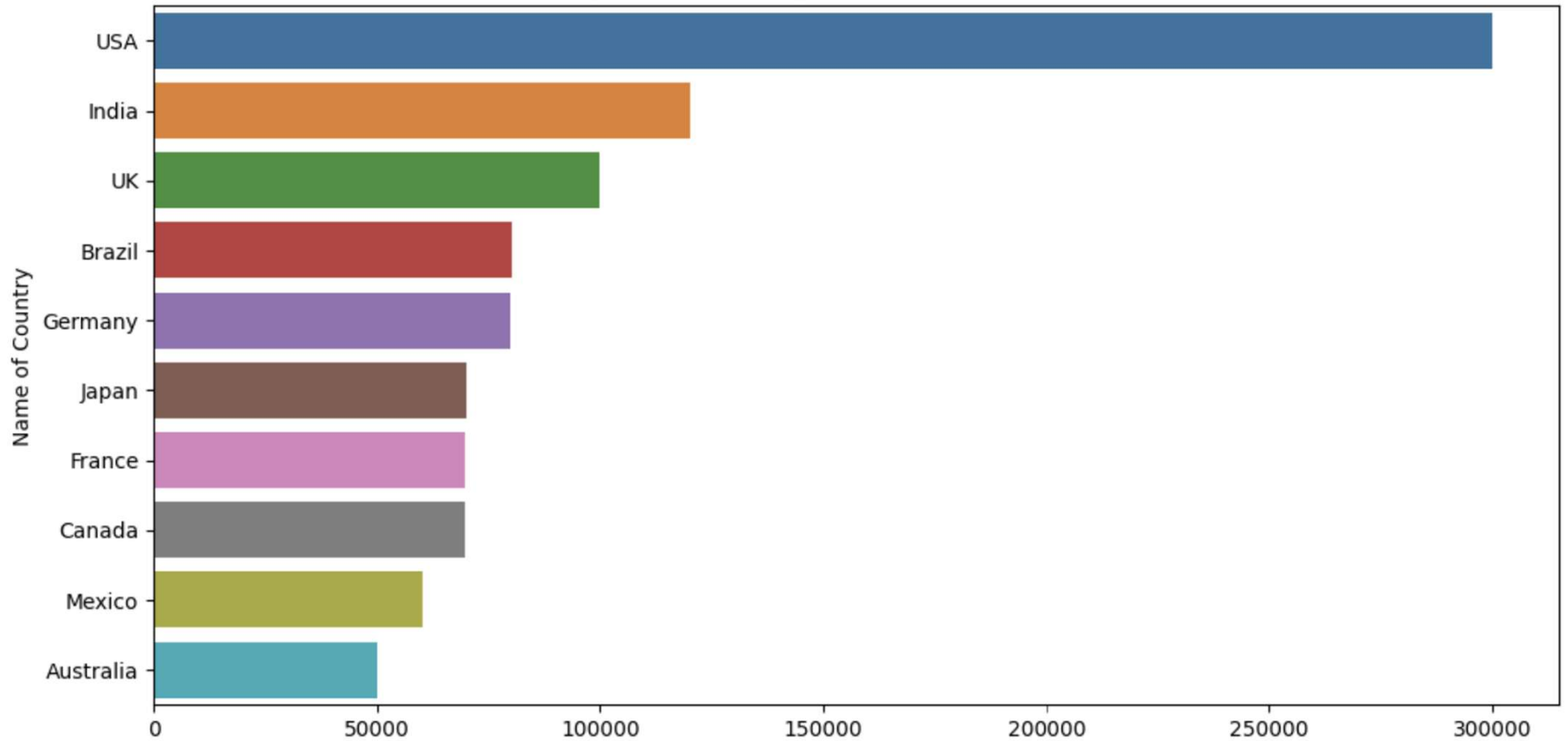
Fraudulent transaction indicator

Type of fraud

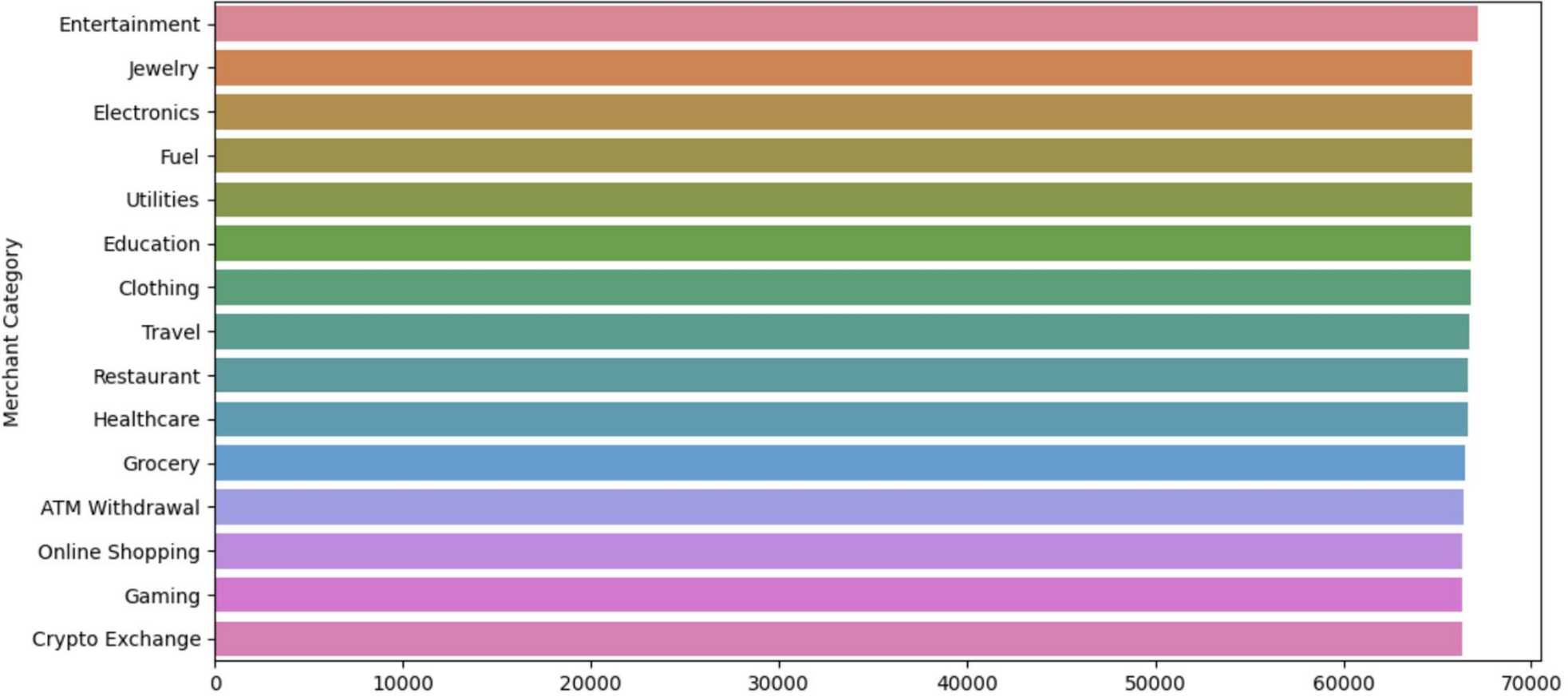
Exploratory Data Analysis

The data set was very clean, had no nulls in any column, and was appropriately formatted. The following slides are several visuals of the data.

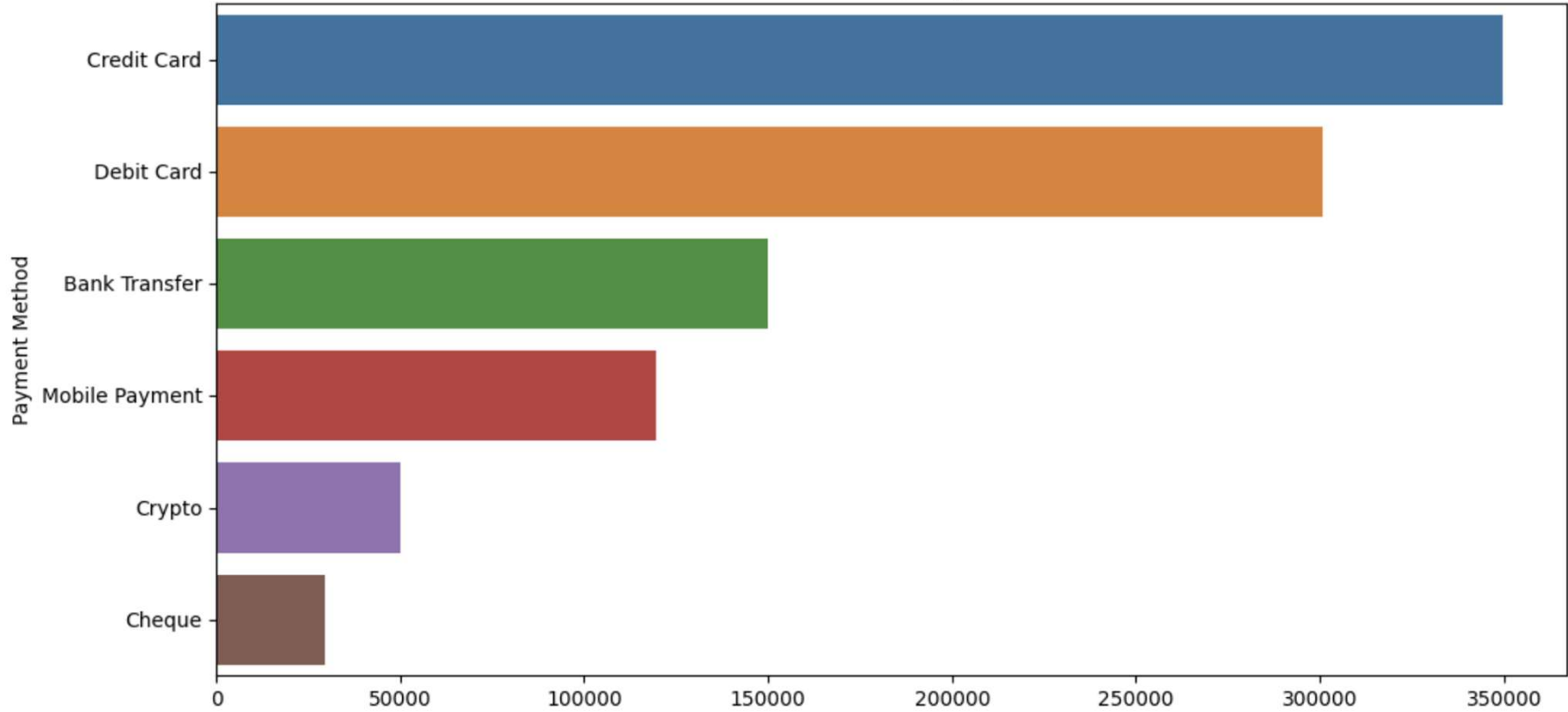
Top 10 Countries by Overall Transation Volume



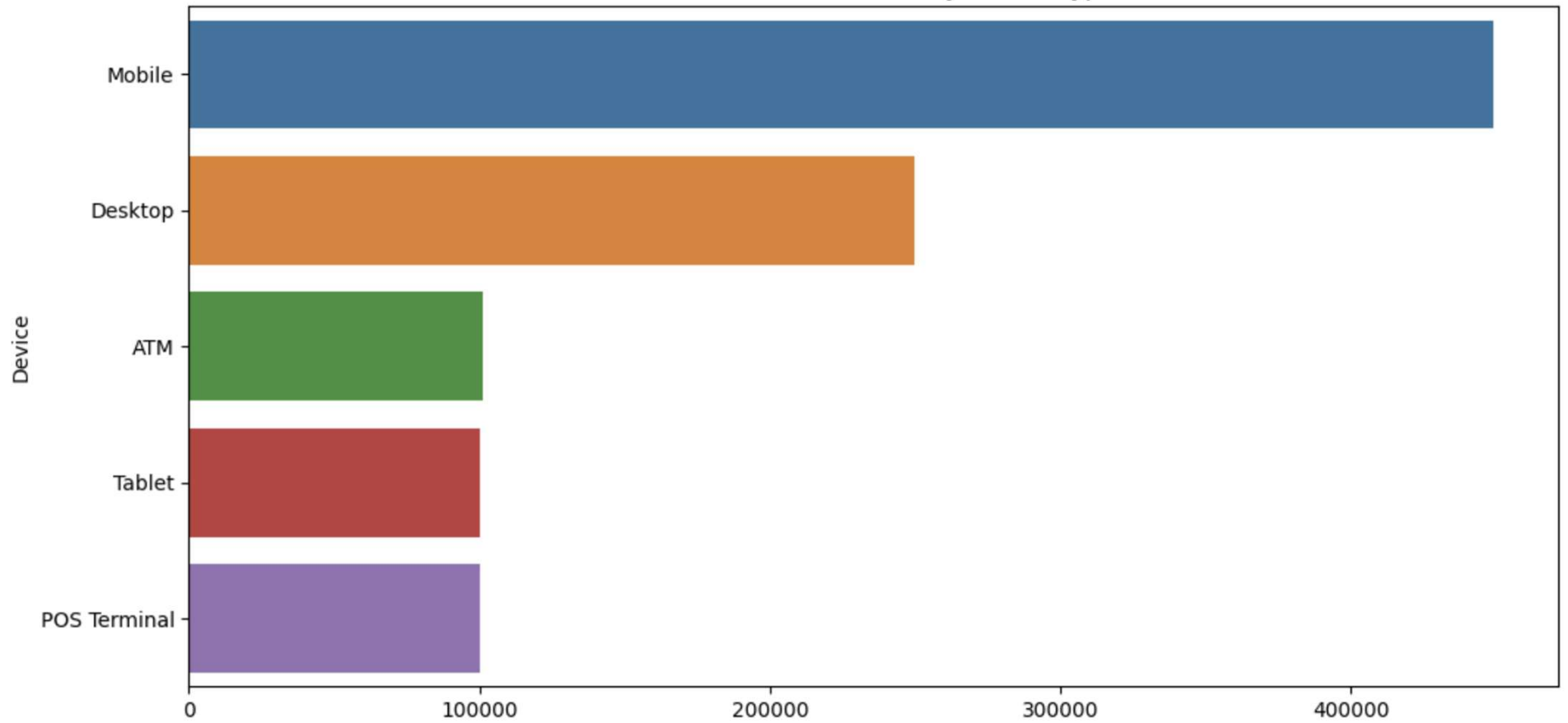
Merchant Category Distribution



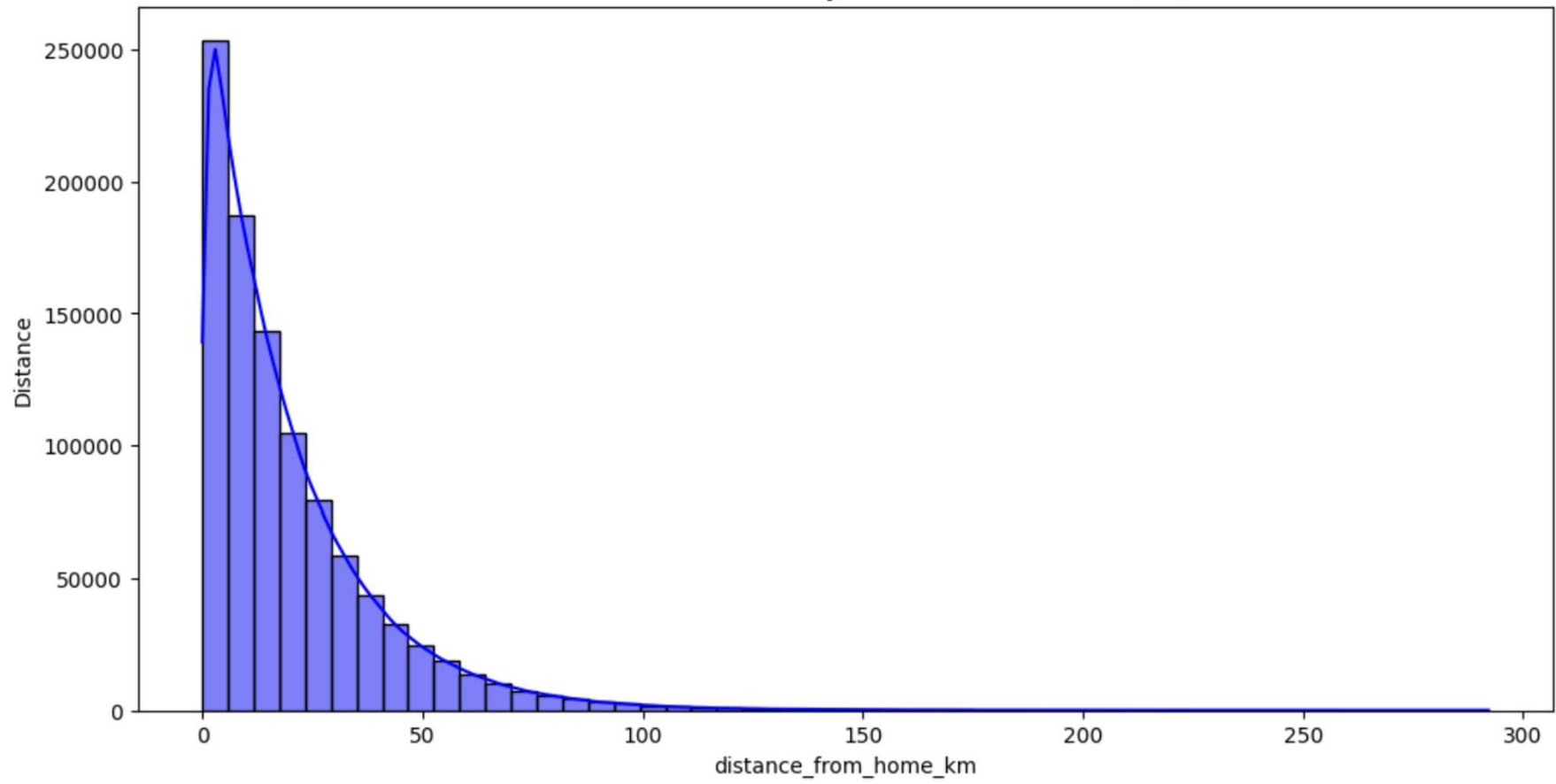
Transaction Count by Payment Method

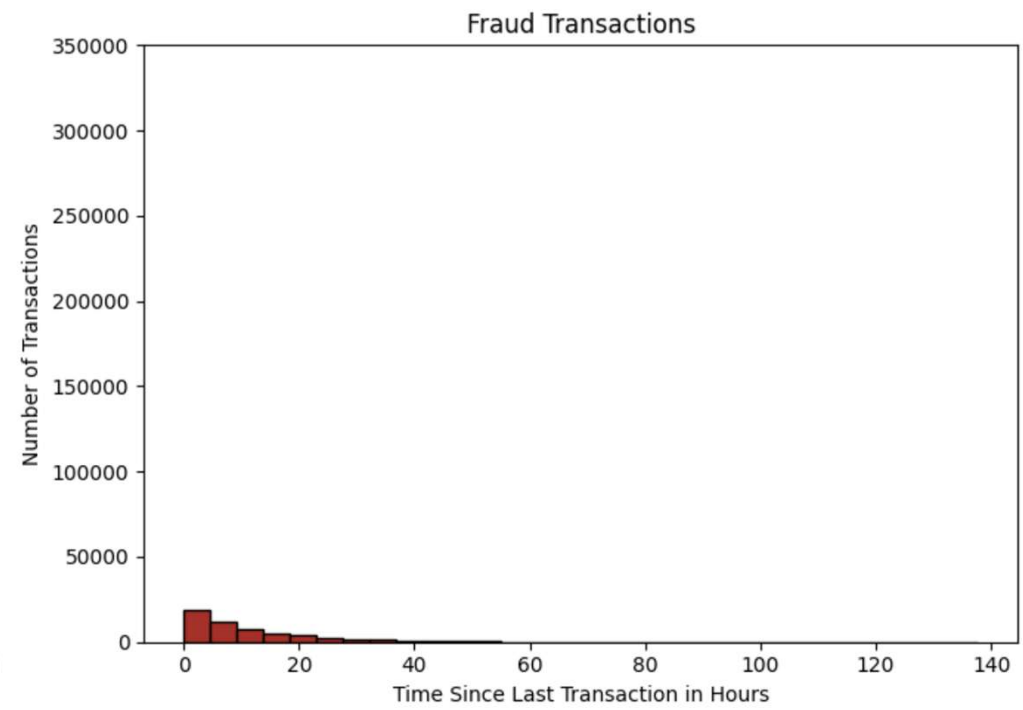
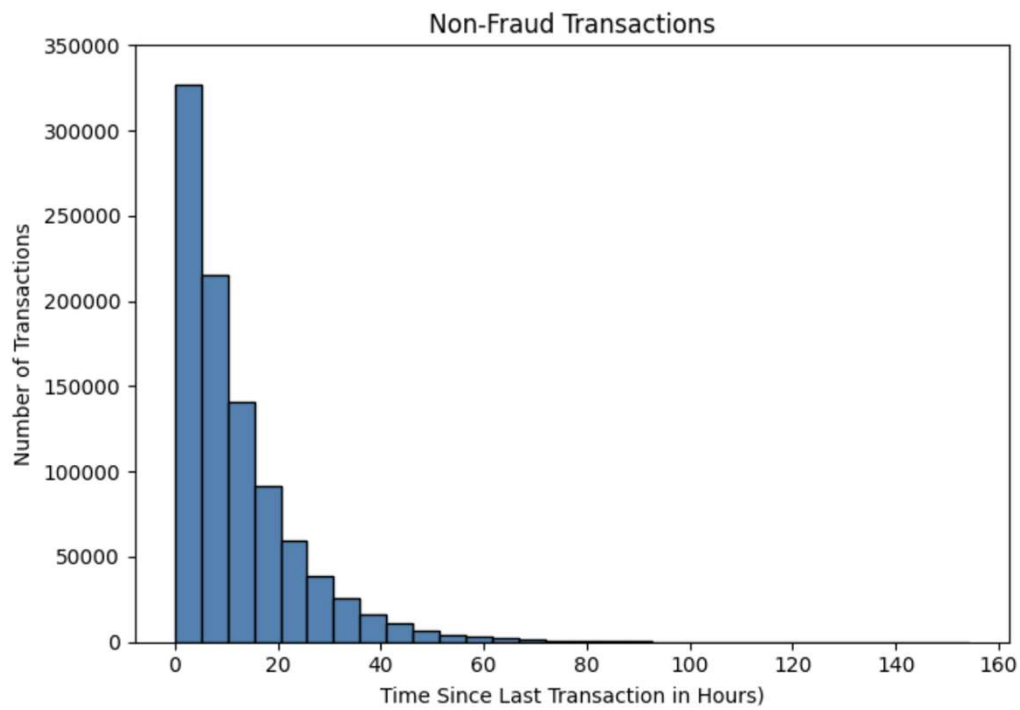


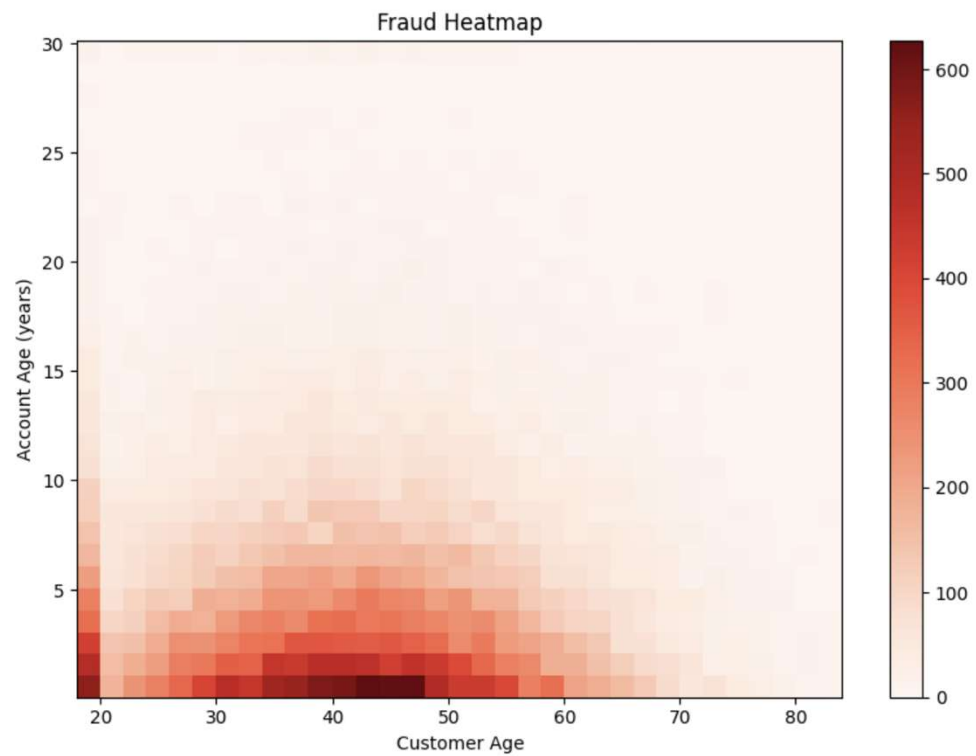
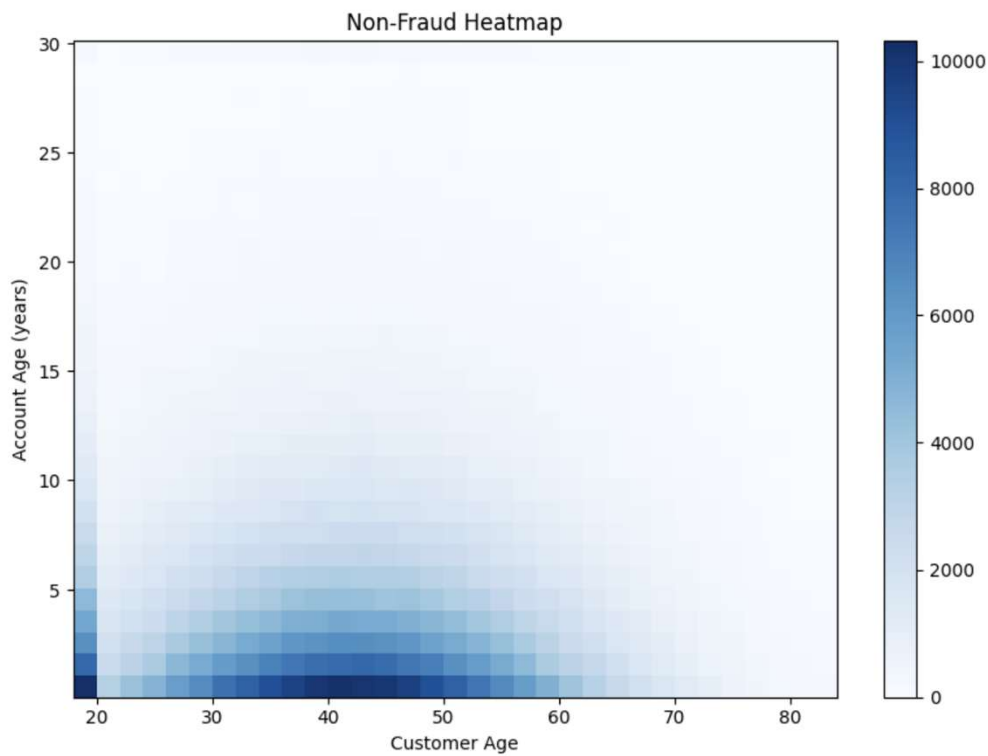
Number of Transactions by Device Type

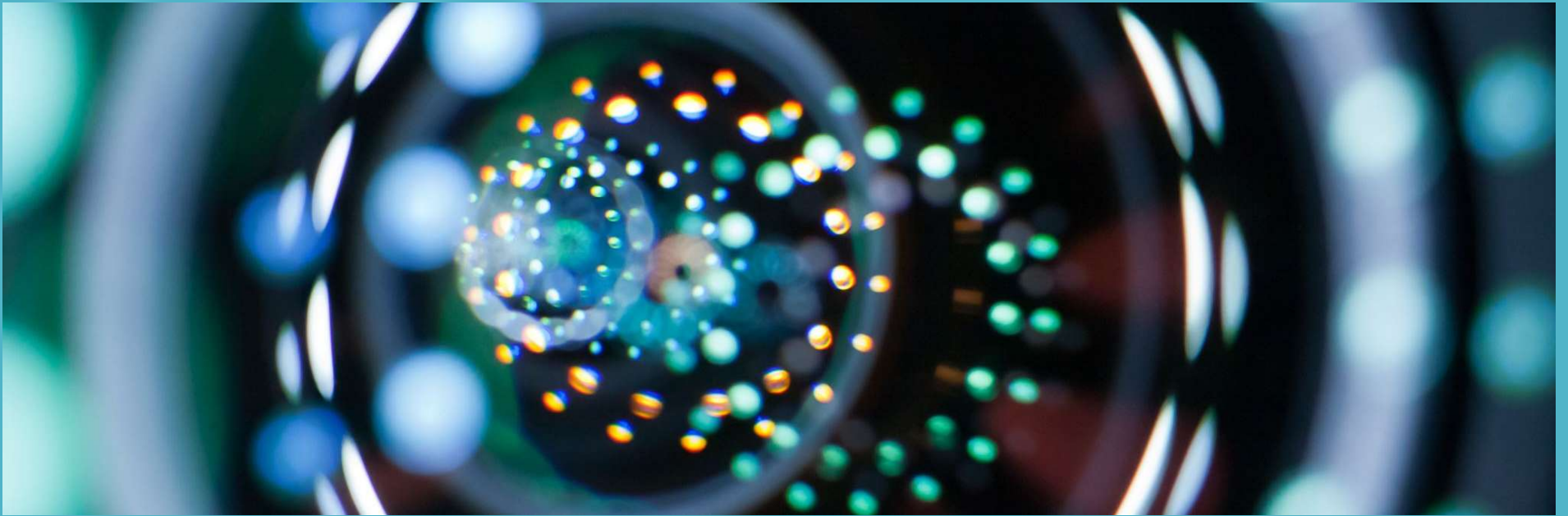


Number of Transactions by Distancce from Home (KM)









Model Creation & Evaluation

Logistic Regression



- Before modeling columns that the model cannot evaluate such as ID, date/time, country, and city were removed
- Categorical fields were one hot encoded
- Data was split the data into test and train groups with the test size of 20% of the rows
- A logistic regression model was ran with:
 - A balanced class weight
 - The liblinear solver
 - A standard scaler
 - Max iterations of 5,000

Model Performance

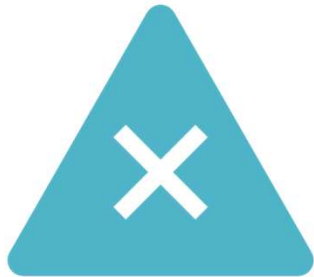
98.42% accuracy

86% precision

83% recall

The false positives
are 1,374 and
false negatives are
1,795.

Conclusion



False positives have more of an impact on the business than the customer



False negatives show that some fraud will not be immediately identified.

Assumptions

People executing the fraud maintain their current strategies and do not adapt to how businesses are approving or declining transactions

Limitations

As the dataset used is composed of synthetic data it may come to be that it is not relevant to the real world.

Challenges

Not having access to the full bank data to identify columns to include in the model may limit its performance



Recommendations

FinCEN data shows that credit and debit card fraud has steadily intensified over the past five years indicating that institutions need a fraud mitigation strategy.



Implementation Plan

Models can be created for implementation into their operational system. Also, consumers need to be educated on what to do if they have unauthorized transactions, which is to contact their bank immediately.

Ethical Assessment

Ethical considerations need to be made for:

Geographic areas that come up during analysis that currently have heavy political issues or wars going on

Bias and fairness in automated decision making through use of AI models

False positives that occur can cause consumer harm and consequences of misclassification

Data privacy and security

Fairness, transparency, consumer protection, responsible data use, and continuous monitoring to ensure that fraud mitigation strategies serve both institutional needs and the well-being of the consumers they are designed to protect.

References

- Banking Fraud Detection. (2024). Transunion.com. <https://www.transunion.com/business-needs/fraud-prevention/banking-fraud-detection>
- Barocas, S., & Selbst, A. D. (2016). *Big data's disparate impact*. California Law Review, 104(3), 671–732.
- Caporal, J. (2026, January 12). *Identity Theft and Credit Card Fraud Statistics for 2025*. The Motley Fool. <https://www.fool.com/money/research/identity-theft-credit-card-fraud-statistics>
- Consumer Financial Protection Bureau. (2022). *CFPB issues guidance to protect consumers from discriminatory practices in automated systems*. <https://www.consumerfinance.gov/>
- Credit Card and Other Financial Instrument Fraud. (2024.). https://www.ussc.gov/sites/default/files/pdf/research-and-publications/quick-facts/Credit_Card_Fraud_FY24.pdf
- Credit Card and Debit Card Fraud | OCC. (2025). OCC.gov. <https://www.occ.gov/topics/consumers-and-communities/consumer-protection/fraud-resources/credit-card-and-debit-card-fraud.html>
- FDIC Information and Support Center. (2026). Fdic.gov. https://ask.fdic.gov/fdicinformationandsupportcenter/s/article/Q-What-should-I-do-if-I-have-unauthorized-charges-on-my-debit-card?language=en_US
- Federal Trade Commission. (2023). *Data security guidance for businesses*. <https://www.ftc.gov/>
- Howell, C. J., & Maimon, D. (2022, December 2). *Darknet Markets Generate Millions in Revenue Selling Stolen Personal data, Supply Chain Study Finds*. The Conversation. <https://theconversation.com/darknet-markets-generate-millions-in-revenue-selling-stolen-personal-data-supply-chain-study-finds-193506>
- IEEE Computational Intelligence Society. (2019). *IEEE-CIS Fraud Detection* [Review of *IEEE-CIS Fraud Detection*]. <https://www.kaggle.com/competitions/ieee-fraud-detection/overview>
- Islam, N. (2026, May 26). *Bank Transaction Fraud Detection Dataset* [Review of *Bank Transaction Fraud Detection Dataset*]. <https://www.kaggle.com/datasets/nafiulislam490/bank-transaction-fraud-detection-dataset>
- Maimon, D. D. (2025, October 1). *Fraud In America 2025: The Consumer Threat Landscape*. Forbes. <https://www.forbes.com/sites/davidmaimon/2025/10/01/fraud-in-america-2025-the-consumer-threat-landscape/>